

**WEST BENGAL STATE ELECTRICITY DISTRIBUTION
COMPANY LIMITED**

(A Govt. of West Bengal Enterprise)

Office of the Chief Engineer,
Procurement & Contracts Department
Vidyut Bhavan, 4th. floor,
Bidhannagar, Kolkata – 700091.

**Technical Specification
for**

**1.1 KV PVC Insulated PVC Sheathed
Multicore Aluminium Cable
(Armoured & Unarmoured)**

Technical specification for 1.1 KV PVC insulated PVC Sheathed Multicore Aluminium Cable (Armoured & Anarmoured).

1.0 SCOPE :

This specification covers design, manufacture, testing at manufacturer's works before despatch, transportation, supply and delivery of 1.1 KV grade PVC insulated PVC sheathed multicore Aluminium Cable.

2.0 LOCATION :

A. 1.1 KV grade 2 Core 4 sq. mm. & 6 sq. mm. PVC unarmoured Cables

The Cable are used for overhead service connection purpose under working voltage up to & including 1100 Volts suitable for LV side of distribution Transformer, any other LV use in 3 Phase earthed system.

B. 1.1 KV grade 4 Core PVC armoured Cables of all sizes

The Cable shall be laid/buried directly in ground anywhere in West Bengal and terminated for outdoor connection.

The Cables may be laid within covered Cable Trenches in Cable Racks/Ladder/open Air etc. for certain portion of length.

The above mentioned cables will be used within the State of West Bengal under WBSEDCL area.

3.0 STANDARD

The materials covered under this specification shall comply with the requirements of the latest version of the following standards as amended up to date, except where specified otherwise.

i)	IS:8130-1984 (amended upto date)	:	Specification for Conductors for insulated electric cables and flexible cords
ii)	IS:5831-1984 (amended upto date)	:	Specification for PVC insulation & sheath of electric cables
iii)	IS : 1554(Part-I) : 1988 (amended upto date)	:	Specification for PVC Insulated heavy duty Electric Cables for working Voltage upto & including 1100 Volt
iv)	IS: 3961(Part-II)-1967 (amended upto date)	:	Recommended current rating for PVC insulated and PVC sheathed heavy duty Cables
v)	IS: 3975-1970 (amended upto date)	:	Specification for Low Carbon Galvanized steel wires, Formed Wires and tapes for armouring of Cables.
vi)	IS:10810-1984 (amended upto date)	:	Methods of test for Cables.
vii)	IS:10418-1982 (amended upto date)	:	Specification for Drums for Electric Cables.

CLIMATIC AND ISOCERAUNIC CONDITIONS:

4.0

Maximum ambient temperature - 50°C

Minimum ambient temperature - 4°C

Thermal resistance of soil - 150°C -cm/watt

Maximum daily average ambient temperature: 45°C

Maximum relative humidity- 100%

Average rainfall per annum-200 cm

Maximum height above the sea level -1000 meter

Maximum wind pressure/wind speed: 150 kg/sq. m/50m/sec

Average no. of thunderstorm days per annum: 200

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

5.0 RATED VOLTAGE:

The rated voltage of the cables shall be 1100 V.

6.0 CABLE SIZE & DATA

A) 1.1 KV grade un-armoured PVC insulated PVC sheathed 2 Core Aluminium Cable

Sl. No	Item Description	Aluminium Conductor	Minimum No. of Strand	Maximum DC Resistance of Conductor at 20°C (Ohm/KM)	Nominal thickness of Insulation	Minimum thickness of inner sheath	Nominal thickness of outer sheath	Minimum thickness of outer sheath	Continuous Current Carrying capacity in air at 40°C ambient	Cable Code
1	1.1 KV PVC Cable 2C X 4 sq. mm	Stranded Circular (non-compacted), H4 grade with flexibility class 2	3	7.41	1.0 mm	0.3	1.8	1.24	27A	AYY
2	1.1 KV PVC Cable 2C X 6 sq. mm		3	4.61	1.0 mm	0.3	1.8	1.24	35A	AYY

B) 1.1 KV Grade armoured PVC insulated PVC sheathed 4 Core Aluminium Cable

Cable size	Aluminium Conductor	Minimum No. of Strand	Maximum DC Resistance of Conductor at 20°C (Ohm/KM)	Nominal thickness of Insulation	Minimum thickness of inner sheath	Armour	Nominal dimensions of Armour With Tolerance	Maximum DC resistance of Armour of Cable at 20°C (Ohm/KM)	Minimum thickness of outer sheath	Continuous Current Carrying capacity in air at 40°C ambient	Continuous Current Carrying capacity in ground at 30°C	Cable Code
1.1 KV PVC Cable 4CX 10 sq. mm	Stranded non-compacted Circular, H4 grade with flexibility class 2	7	3.08	1.0	0.30	Galvanized steel Strip	4 X 3.4 X 0.8 mm (±10%)	4.68	1.40	40A	46A	AYFY
1.1 KV PVC Cable 4C X 25 sq. mm	Stranded compacted Circular or shaped, H4 grade with flexibility class 2	6	1.20	1.2	0.30	Galvanized steel Strip	4 X 3.4 X 0.8 mm (±10%)	3.53	1.40	70A	76A	AYFY
1.1 KV PVC Cable 4CX 50 sq. mm		6	0.641	1.4	0.40	Galvanized steel Strip	4 X 3.4 X 0.8 mm (±10%)	2.92	1.56	105 A	110 A	AYFY
1.1 KV PVC Cable 4CX 120 sq. mm		15	0.253	1.6	0.50	Galvanized steel Strip	4 X 3.4 X 0.8 mm (±10%)	2.01	1.88	180 A	185 A	AYFY
1.1 KV PVC Cable 4C X 185 sq. mm		30	0.164	2.0	0.60	Galvanized steel Strip	4 X 3.4 X 0.8 mm (±10%)	1.64	2.04	240 A	235 A	AYFY

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

1.1 KV PVC Cable 4CX300 sq. mm	30	0.100	2.4	0.70	Galvani zed steel Strip	4 X 3.4 X 0.8 mm (±10%)	1.30	2.52	315 A	305 A	AYFY
1.1 KV PVC Cable 4CX 400 sq. mm	53	0.0778	2.6	0.70	Galvani zed steel Strip	4 X 3.4 X 0.8 mm (±10%)	1.20	2.84	375 A	335 A	AYFY

7.0 CONDUCTORS :

Aluminium conductors shall be H4 grade with flexibility class 2 complying with the requirements as specified in IS-8130-1984 with latest amendments.

The Conductors shall be Stranded Circular (non-compacted) for Cross Section of 4 sq. mm., 6 sq. mm. & 10 sq. mm. with flexibility class 2.

Conductors shall be Stranded compacted Circular or shaped conductor for Cross Section above 10 sq. mm. with flexibility class 2.

The conductor shall be clean & reasonably uniform in size and shape and its surface shall be free from sharp edges.

8.0 INSULATION :

The Conductor shall be provided with PVC insulation applied by extrusion. Type of insulation shall be A for general purpose PVC. Maximum permissible Conductor temperature should be 70°C under normal continuous operation & should be 160°C under short circuit condition.

9.0 LAYING UP OF CORES:

In twin, three & multi core cables, the cores shall be laid up together with a suitable lay, the outermost layer shall have right hand lay and the successive layer shall be laid with opposite lay.

10.0 INNER SHEATH:

The laid up cores shall be provided with Inner Sheath applied by extrusion. It shall be ensured that the shape of Cable is as circular as possible. Inner sheath shall be so applied that it fits closely on the laid up cores and it shall be possible to remove it without damaging the insulation.

For Cables with general purpose insulation, the inner sheath shall be of type ST1 PVC Compound conforming to the requirements of IS: 5831-1984 with latest amendments.

11.0 ARMOURING (for Armoured Cable only):

The armour wires/strips shall be applied as closely as practicable. A binder tape may be provided on the armour.

The joints in armour wire or strip shall be made by brazing or welding and the surface irregularities shall be removed. A joint in any wire/strip shall be at least 300 mm from the nearest joint in any other armour wire/flat strip in the completed cable.

Where the calculated dia. below armouring does not exceed 13 mm, the armour shall consist of round wires. Where the calculated dia. below armouring exceeds 13 mm, the armour shall consist of flat strips.

For flat strips armour nominal dimensions of armour should be 4 mm. X 3.4 mm. X 0.8 mm(±10%).

Armour Coverage Percentage for cables shall not be less than 90% as per IS: 1554: (Part-1) 2012 with its Amendment no. 5 April 2012.

12.0 OUTER SHEATH :

The outer sheath shall be applied by extrusion over the inner sheath in case of unarmoured cable & over the armouring in case of armoured Cable. The colour of outer sheath is black unless otherwise specified. Thickness of outer sheath should be as per IS: 1554 (Part-I): 1988 with latest

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

amendment.

For Cables with general purpose insulation, the outer sheath shall be of type ST1 PVC Compound conforming to the requirements of IS:5831-1984 with latest amendments.

13.0 CLASSIFICATION OF TESTS

Type Test Report

During bid submission, the bidder shall submit Type Test report for each offered item.

Type Test should be carried out

- i) Within 5 years from the due date of opening of tender.
- ii) From CPRI or ERDA or NABL accredited LAB bearing NABL Logo. Accreditation of NABL LAB should be displayed in the official website of NABL.
- iii) As per IS: 1554(Part-1)-1988 and its latest amendment and other relevant IS.

All the tests mentioned below are to be made as per details given in IS: 10810-1984. The following shall constitute Type Test:

- a) Resistance Test on Conductor
- b) Tests for armouring Wires/strips (for Armoured Cable).
- c) Test for thickness of insulation and sheath
- d) Physical tests for insulation and outer sheath
- i) Tensile strength and elongation at break
- ii) Ageing in air oven
- iii) Shrinkage test
- iv) Hot deformation
- v) Loss of mass in air oven
- vi) Heat shock test
- vii) Thermal stability test
- e) Insulation resistance Test
- f) High voltage test (water immersion test)
- g) High voltage test at room temperature
- h) Flammability test

ACCEPTANCE TEST:

The following shall constitute acceptance Test:

- a) Resistance Test on Conductor
- b) Test for thickness of insulation and sheath
- c) Tensile strength and elongation at break test of insulation & sheath.
- d) Insulation resistance Test
- e) High voltage test at room temperature.

ROUTINE TEST:

The following shall constitute routine test:

- a) Resistance Test on Conductor
- b) High voltage test at room temperature

14.0 CABLE IDENTIFICATION & SEQUENTIAL LENGTH MARKING:

Cable identification by embossing:

The following shall be embossed on the outer sheath of the cable throughout the length of cable at 1.0 meter spacing for identification:

- i) Manufacturer's Name or Trade Mark.
- ii) Type of Cable with Cable code
- iii) Voltage Grade of Cable (1100 V)

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

- iii) Type of insulation, material of conductor (PVC insulated Aluminium conductor)
- iv) Number of cores & nominal cross sectional area of conductor
- v) Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)
- vi) Cable Code
- vii) Marking "Electric-WBSEDCL".
- viii) Month & Year of manufacture

Sequential length marking by printing/ engraving

Sequential length shall be marked on the outer sheath of the cable throughout the length by Printing in each meter length interval.

15.0 PACKING & DRUM DETAILS:

The Cable shall be supplied in non-returnable wooden drums having drum length as mentioned below with tolerance of $\pm 5\%$ per drum subject to overall maximum minus (-) 1% on total delivered quantity will be accepted.

Cable Size	Drum Length
A) 1.1 KV grade unarmoured PVC insulated PVC sheathed 2 Core Aluminium Cable	
1.1 KV PVC Cable 2 Core 4 sq. mm	500 ± 5%
1.1 KV PVC Cable 2 Core 6 sq. mm	
B) 1.1 KV grade armoured PVC insulated PVC sheathed 4 Core Aluminium Cable	
1.1 KV PVC Cable 4 Core 10 sq. mm	500 ± 5%
1.1 KV PVC Cable 4 Core 25 sq. mm	
1.1 KV PVC Cable 4 Core 50 sq. mm	
1.1 KV PVC Cable 4 Core 120 sq. mm	250± 5%
1.1 KV PVC Cable 4 Core 185 sq. mm	
1.1 KV PVC Cable 4 Core 300 sq. mm	
1.1 KV PVC Cable 4 Core 400 sq. mm	

The drums shall be proofed against attack by white ant or termite, conforming to IS:10418:1982. The clearance between the outermost layer of the cable & the edge of the flange should be at least 75 mm or equal to the diameter of the Cable whichever is less.

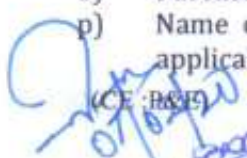
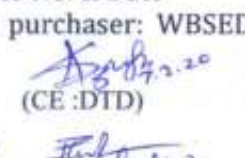
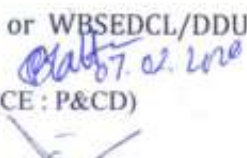
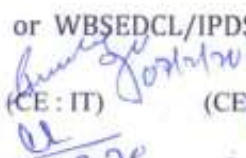
16.0 SEALING OF CABLE ENDS:

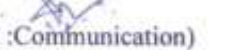
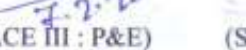
The ends of the cable shall be sealed with heat shrinkable caps.

17.0 MARKING ON EACH DRUM

The following information shall be stenciled on each drum:

- a) Reference to this Indian Standard, for example, Ref IS:1554(Part-I)
- b) Manufacturer's Name, brand name or Trade Mark
- c) License Number
- d) Drum Identification No.
- e) Type of Cable and Voltage Grade
Example - 1.1 KV grade PVC insulated PVC sheathed Aluminium Cable (armoured/unarmoured)
- f) Number of cores
- g) Nominal cross sectional area of Conductor
- h) Cable Code
- i) Length of Cable on the Drum
- j) Number of length on the drum(if more than one)
- k) Direction of rotation of Drum by means of arrow.
- l) Approximate gross weight & tare weight
- m) Month & Year of manufacture
- n) Month of Delivery
- o) Purchase Order No. & Date
- p) Name of the purchaser: WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS(as applicable).

 (ACE : Project III)
  (ACE : DTD)
  (ACE : P&CD)
  (CE : IT)
  (CE : CCD)

 (ACE : Project III)
  (ACE : DTD)
  (ACE : Communication)
  (ACE III : P&E)
  (SE : P&E)

GUARANTEED TECHNICAL PARTICULARS

for

1.1 KV Grade PVC Cable of size 2Cx4 sq. mm. & 2Cx6 sq. mm.

(To be filled in by the Supplier)

Sl. No.	Name of the Article	Guaranteed Particulars	
		1.1 KV 2 Core 4 sq. mm	1.1 KV 2 Core 6 sq. mm.
	Name of the Manufacturer		
	Factory Address		
1	CABLE Code	AYY	AYY
2	RATED VOLTAGE	Working Voltage up to and including 1100 Voltage	Working Voltage up to and including 1100 Voltage
3	Applicable Standard	IS:8130-1984, IS:5831-1984, IS : 1554(Part-I) : 1988, IS: 3961(Part-II)-1967, IS:10810-1984 & IS:10418-1982 - All the IS with latest amendments	
4	Conductor		
i)	Material	H4 Grade Aluminium wires with flexibility Class 2 as per IS: 8130	H4 Grade Aluminium wires with flexibility Class 2 as per IS: 8130
ii)	Shape of Conductor	Stranded non-compacted circular	Stranded non-compacted circular
iii)	Maximum DC Resistance of Conductor at 20°C (Ohm/KM)	7.41	4.61
iv)	Nominal Area of Cross Section (sq. mm)	4 sq. mm	6 sq. mm
v)	No. of Strand		
vi)	Strand Dia. in mm		
5	Insulation		
i)	Material	Extruded PVC Type 'A' of IS: 5831-84 with latest amendment	Extruded PVC Type 'A' of IS: 5831-84 with latest amendment
ii)	Nominal thickness of Insulation	1.0 mm	1.0 mm
iii)	Suitability with regard to temp, moisture, Acid, Oil & Alkaline Surrounding	Yes	Yes
6.	Inner Sheath		
i)	Material	Extruded PVC type (ST-1) as per IS: 5831 : 84 with latest Amendment	Extruded PVC type (ST-1) as per IS: 5831 : 84 with latest Amendment
ii)	Minimum thickness of inner sheath (mm)	0.30 mm	0.30 mm
7	Outer sheath		
i)	Material	Extruded Black PVC type (ST-1) as per IS: 5831 : 84 with latest Amendment	Extruded Black PVC type (ST-1) as per IS: 5831 : 84 with latest Amendment
ii)	Minimum thickness of outer sheath	1.24 mm	1.24 mm
iii)	Nominal thickness of outer sheath (mm)	1.80 mm	1.80 mm
8	Overall minimum dia of Cable (mm)	10.88 mm	11.88 mm.
9	Method of Core identification	As PER IS:1554 (Part-I) 1988 with latest amendments	As PER IS:1554 (Part-I) 1988 with latest amendments
10	Maximum operating temperature of Conductor	70°C	
11	Maximum temperature of	160°C	

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

	conductor during short circuit																		
12	Drum length & tolerance of each drum	500±5%																	
13	Overall tolerance in total quantity	-1%																	
14	Continuous Current Carrying capacity in air at 40°C	27 A	35 A																
15	Sealing of cable ends	The ends of the cable shall be sealed with heat shrinkable caps																	
16	Approximate net weight of Cable (Kg./KM)																		
17	Cable identification & sequential length marking	Cable identification by Embossing The following shall be embossed on the outer sheath of the cable throughout the length of cable at 1.0 meter spacing for identification: <table><tr><td>i)</td><td>Manufacturer's Name or Trade Mark</td></tr><tr><td>ii)</td><td>Type of cable with Cable Code</td></tr><tr><td>iii)</td><td>Voltage Grade of Cable</td></tr><tr><td>iv)</td><td>Type of insulation, material of conductor (PVC insulated Aluminium conductor)</td></tr><tr><td>v)</td><td>Number of cores & Nominal cross sectional area of conductor</td></tr><tr><td>vi)</td><td>Month & Year of manufacture</td></tr><tr><td>vii)</td><td>Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)</td></tr><tr><td>viii)</td><td>Marking "Electric WBSEDCL".</td></tr></table> Sequential length marking by printing/ engraving Sequential length shall be marked on the outer sheath of the cable throughout the length by Printing/ engraving in each meter length interval.		i)	Manufacturer's Name or Trade Mark	ii)	Type of cable with Cable Code	iii)	Voltage Grade of Cable	iv)	Type of insulation, material of conductor (PVC insulated Aluminium conductor)	v)	Number of cores & Nominal cross sectional area of conductor	vi)	Month & Year of manufacture	vii)	Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)	viii)	Marking "Electric WBSEDCL".
i)	Manufacturer's Name or Trade Mark																		
ii)	Type of cable with Cable Code																		
iii)	Voltage Grade of Cable																		
iv)	Type of insulation, material of conductor (PVC insulated Aluminium conductor)																		
v)	Number of cores & Nominal cross sectional area of conductor																		
vi)	Month & Year of manufacture																		
vii)	Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)																		
viii)	Marking "Electric WBSEDCL".																		
18	Marking on each drum	The following information shall be stenciled on each drum: a) Indian Standard, for example, Ref IS:1554(Part-I) b) Manufacturer's Name, brand name or Trade Mark c) License Number. d) Drum Identification No. e) Type of Cable and Voltage Grade Example - 1.1 KV grade PVC insulated PVC sheathed Aluminium Cable (armoured/unarmoured) f) Number of cores g) Nominal cross sectional area of Conductor h) Cable Code i) Length of Cable on the Drum j) Number of length on the drum(if more than one) k) Direction of rotation of Drum by means of arrow. l) Approximate gross weight & tare weight m) Month & Year of manufacture n) Month of Delivery o) Purchase Order No. & Date p) Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable).																	

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

GUARANTEED TECHNICAL PARTICULARS

for

1.1 KV Grade 4Core PVC Cable of size 10 sq. mm., 25 sq. mm, 50 sq. mm, 120 sq. mm, 185 sq. mm, 300 sq. mm. & 400 sq. mm

(To be filled in by the Supplier)

Sl. No.	Name of the Article	Guaranteed Particulars						
		4CX10 sq. mm	4CX25 sq. mm	4CX50 sq. mm	4CX120 sq. mm	4CX185 sq. mm	4CX300 sq. mm	4CX400 sq. mm
	Name of the Manufacturer							
	Factory Address							
1	CABLE Code	AYFY	AYFY	AYFY	AYFY	AYFY	AYFY	AYFY
2	RATED VOLTAGE	Working Voltage up to and including 1100 Voltage						
3	Applicable Standard	IS:8130-1984, IS:5831-1984, IS : 1554(Part-I) : 1988, IS: 3961(Part-II)-1967, IS: 3975-1970, IS:10810-1984 & IS:10418-1982 - All the IS with latest amendments						
4	Conductor							
i)	Material	H4 Grade Aluminium wires with flexibility Class 2 as per IS: 8130						
ii)	Shape of Conductor	Stranded non-Compacted Circular	Stranded Compacted Circular or shaped	Stranded Compacted Circular or shaped	Stranded Compacted Circular or shaped	Stranded Compacted Circular or shaped	Stranded Compacted Circular or shaped	Stranded Compacted Circular or shaped
iii)	Maximum DC Resistance of Conductor at 20°C (Ohm/KM)	3.08	1.20	0.641	0.253	0.164	0.100	0.0778
iv)	Nominal Area of Cross Section (sq. mm)	10	25	50	120	185	300	400
v)	No. of Strand							
vi)	Strand Dia. in mm							
5	Insulation							
i)	Material	Extruded PVC Type 'A' of IS: 5831- 84 with latest amendment						
ii)	Nominal thickness of Insulation	1.0	1.2	1.4	1.6	2.0	2.4	2.6
iii)	Suitability with regard to temp, moisture, Acid, Oil & Alkaline Surrounding	Yes						
6.	Inner Sheath							
i)	Material	Extruded PVC type (ST-1) as per IS: 5831 : 84 with latest Amendment						
ii)	Minimum thickness of inner sheath (mm)	0.30	0.30	0.40	0.50	0.60	0.70	0.70
7	Armour							
i)	Armouring material	Galvanized Steel Strip						
ii)	Nominal dimensions of Armour	4X3.4X0.8 mm (±10%) as per IS:3975-1970 amended upto date						
iii)	Maximum DC resistance of Armour of Cable at 20°C (Ohm/KM)	4.68	3.53	2.92	2.01	1.64	1.30	1.20
iv)	Armour Coverage Percentage	Minimum 90%						
8	Outer sheath							
i)	Material	Extruded Black PVC type (ST-1) as per IS: 5831 : 84 with latest Amendment						
ii)	Minimum thickness of outer sheath	1.40	1.40	1.56	1.88	2.04	2.52	2.84
9	Overall minimum dia of Cable (mm)	17.10	23.14	30.13	42.37	51.97	64.72	73.73
10	Method of Core identification	As PER IS:1554 (Part-I) 1988 with latest amendments						

(CE : P&E)

(CE : DTD)

(CE : P&CD)

(CE : IT)

(CE : CCD)

(ACE : Project III)

(ACE : DTD)

(ACE : Communication)

(ACE III : P&E)

(SE : P&E)

1	Maximum operating temperature of Conductor	70°C																						
12	Maximum temperature of conductor during short circuit	160°C																						
13	Drum length & tolerance of each drum	500±5%	500±5%	500±5%	250±5%	250±5%	250±5%	250±5%																
14	Overall tolerance in total quantity	-1%																						
15	Continuous Current Carrying capacity in air at 40°C	40 A	70 A	105 A	180 A	240 A	315 A	375 A																
16	Continuous Current Carrying capacity in ground at 30°C	46 A	76 A	110 A	185 A	235 A	305 A	335 A																
17	Sealing of cable ends	The ends of the cable shall be sealed with heat shrinkable caps																						
18	Approximate net weight of Cable																							
19	Cable identification & sequential length marking	Cable identification by Embossing The following shall be embossed on the outer sheath of the cable throughout the length of cable at 1.0 meter spacing for identification: <table><tr><td>i)</td><td>Manufacturer's Name or Trade Mark</td></tr><tr><td>ii)</td><td>Type of cable with Cable Code</td></tr><tr><td>iii)</td><td>Voltage Grade of Cable</td></tr><tr><td>iv)</td><td>Type of insulation, material of conductor (PVC insulated Aluminium conductor)</td></tr><tr><td>v)</td><td>Number of cores & Nominal cross sectional area of conductor</td></tr><tr><td>vi)</td><td>Month & Year of manufacture</td></tr><tr><td>vii)</td><td>Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)</td></tr><tr><td>viii)</td><td>Marking "Electric WBSEDCL".</td></tr></table> Sequential length marking by printing/ engraving Sequential length shall be marked on the outer sheath of the cable throughout the length by Printing/ engraving in each meter length interval.							i)	Manufacturer's Name or Trade Mark	ii)	Type of cable with Cable Code	iii)	Voltage Grade of Cable	iv)	Type of insulation, material of conductor (PVC insulated Aluminium conductor)	v)	Number of cores & Nominal cross sectional area of conductor	vi)	Month & Year of manufacture	vii)	Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)	viii)	Marking "Electric WBSEDCL".
i)	Manufacturer's Name or Trade Mark																							
ii)	Type of cable with Cable Code																							
iii)	Voltage Grade of Cable																							
iv)	Type of insulation, material of conductor (PVC insulated Aluminium conductor)																							
v)	Number of cores & Nominal cross sectional area of conductor																							
vi)	Month & Year of manufacture																							
vii)	Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable)																							
viii)	Marking "Electric WBSEDCL".																							
20	Marking on each drum	The following information shall be stenciled on each drum: a) Indian Standard, for example, Ref IS:1554(Part-I) b) Manufacturer's Name, brand name or Trade Mark c) License Number. d) Drum Identification No. e) Type of Cable and Voltage Grade Example - 1.1 KV grade PVC insulated PVC sheathed Aluminium Cable (armoured/unarmoured) f) Number of cores g) Nominal cross sectional area of Conductor h) Cable Code i) Length of Cable on the Drum j) Number of length on the drum(if more than one) k) Direction of rotation of Drum by means of arrow. l) Approximate gross weight & tare weight m) Month & Year of manufacture n) Month of Delivery o) Purchase Order No. & Date p) Name of the purchaser : WBSEDCL or WBSEDCL/DDUGJY or WBSEDCL/IPDS (as applicable).																						

(CE:P&E)

(CE:DTD)

(CE:P&CD)

(CE:IT)

(CE:CCD)

(ACE:Project III)

(ACE:DTD)

(ACE:Communication)

(ACE III:P&E)

(SE:P&E)